



LAB REPORT

Lab No: 01

Lab Title: GitHub Identity, Student Portal Registration, and Ubuntu Server Installation

Name: Iraj Taimur Malik

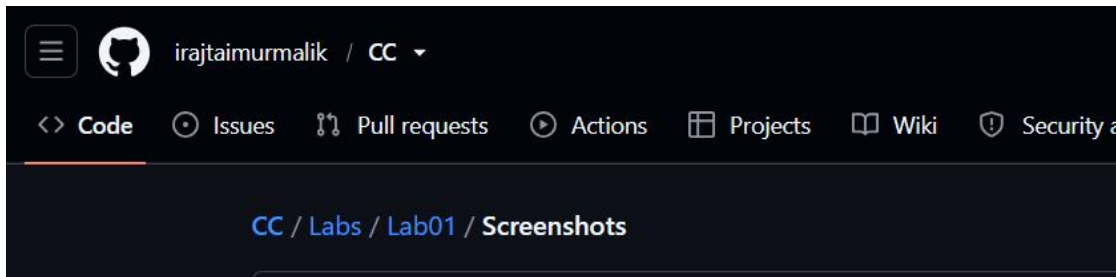
Registration Number: FA24B1-SE-015

Course Code: BSE-665

Section: A

GitHub Username: irajtaimurmalik

Getting Starting:



Task 1: Create and Configure Your GitHub Account

Sign up for GitHub

Email

irajtaimur@gmail.com

[Use a different Google account](#)

Username*

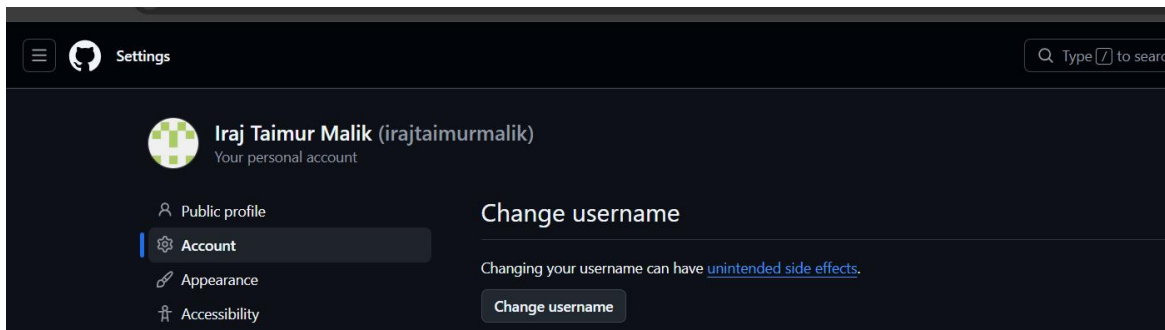
irajtaimurmalik

Your Country/Region*

Pakistan

For compliance reasons, we're required to collect country information to send you occasional updates and announcements.

Caption 1: The GitHub sign-up form filled with the required email address, username, and country of residence.



Caption 2: The GitHub Account settings page, used to update the account username.



Caption 3: The GitHub public profile page displaying the configured display name and public email address.

Task 2: Register in the Student Marks Portal



Caption 4: The landing page of the Student Marks Portal used for registration.

Registration Form

ID

Complete registration

Enter your registration number, course and section.
The enrollment requires approval.

Registration number

2024-BSE-15

Allowed: 2024-BSE-01 to 2024-BSE-99.

Course

SE-665 — Cloud Computing

Section

Section A

Submit enrollment

Sign out

Caption 5: The registration form completed on the Student Marks Portal.

Enrollment Confirmation

STUDENT MARKS PORTAL

Course enrollments

Sign out



Iraj Taimur Malik
2024-BSE-15

+ Enroll in another course

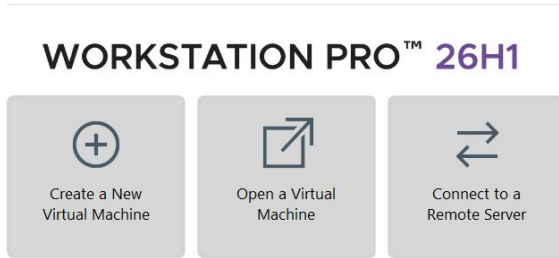
ENROLLMENT STATUS

Your courses

Cloud Computing (SE-665) Section A
Waiting for administrator approval
Pending approval

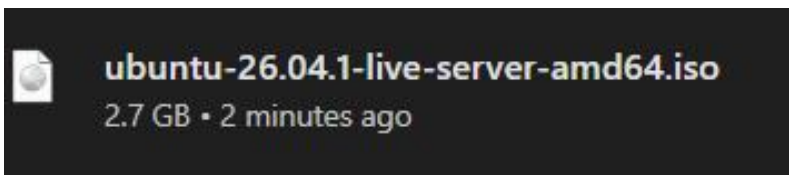
Caption 6: The enrollment page confirming successful registration on the Student Marks Portal.

Task 3: Download VMware Workstation Pro and Ubuntu Server



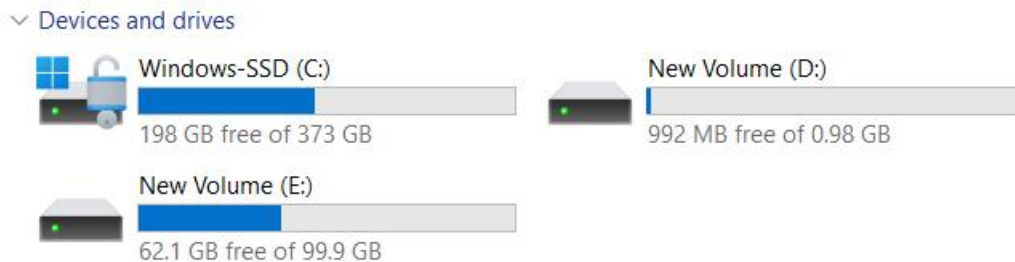
Caption 7: The official download page for VMware Workstation Pro.

Ubuntu Server ISO File



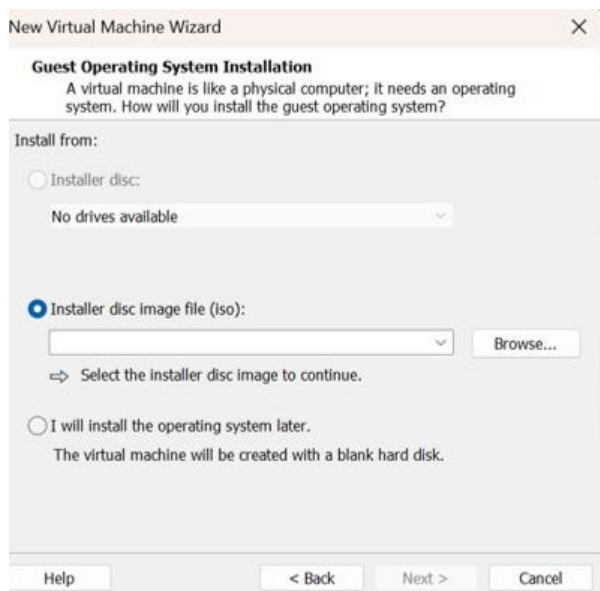
Caption 8: The official download page for the Ubuntu Server ISO image.

Available Disk Storage



Caption 9: Available storage space on the host machine, verified before creating the virtual disk.

Task 4: Create the Ubuntu Server Virtual Machine



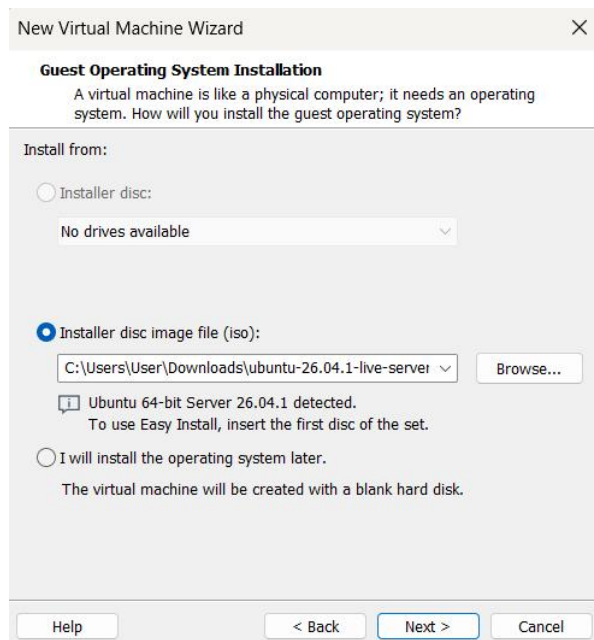
Caption 10: The welcome screen of the New Virtual Machine Wizard in VMware Workstation.

Selecting the Typical Configuration



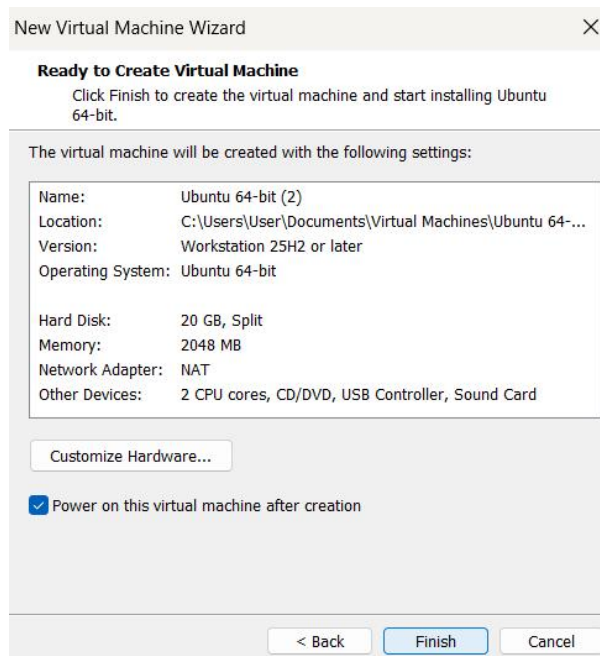
Caption 11: The Typical (recommended) configuration option selected for the new virtual machine.

Selecting the Ubuntu Server ISO File



Caption 12: Browsing to and selecting the downloaded Ubuntu Server ISO file as the installation source.

Virtual Machine Configuration Summary



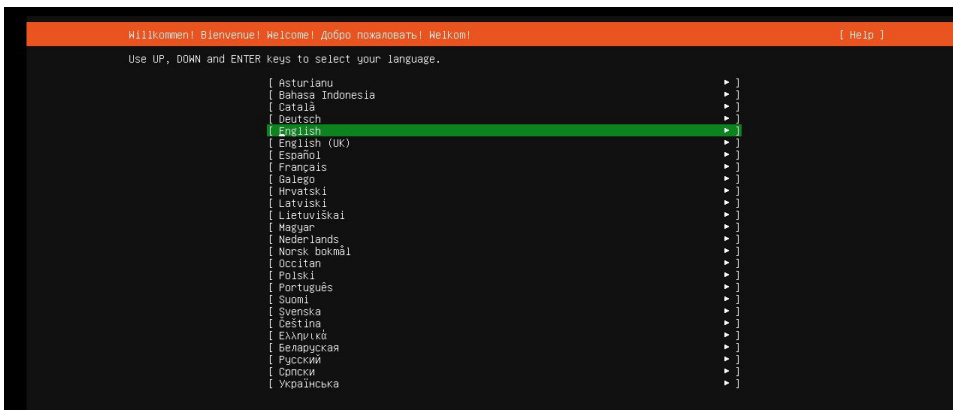
Caption 13: A summary of the virtual machine configuration shown before final creation.

Ubuntu installer boot waiting



Caption 14: The Ubuntu Server installer boot menu waiting to load the installation environment.

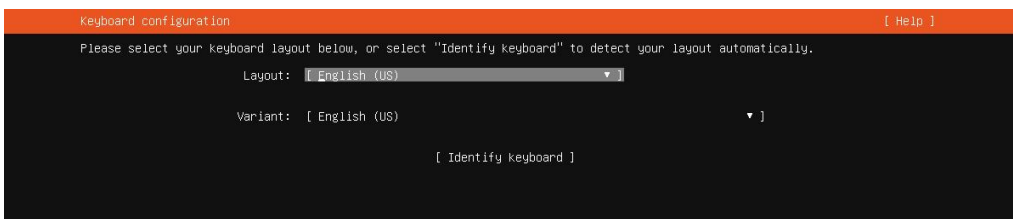
Task 5: Choose the Installation Language



Caption 15: The language selection screen presented at the start of the Ubuntu Server installer.

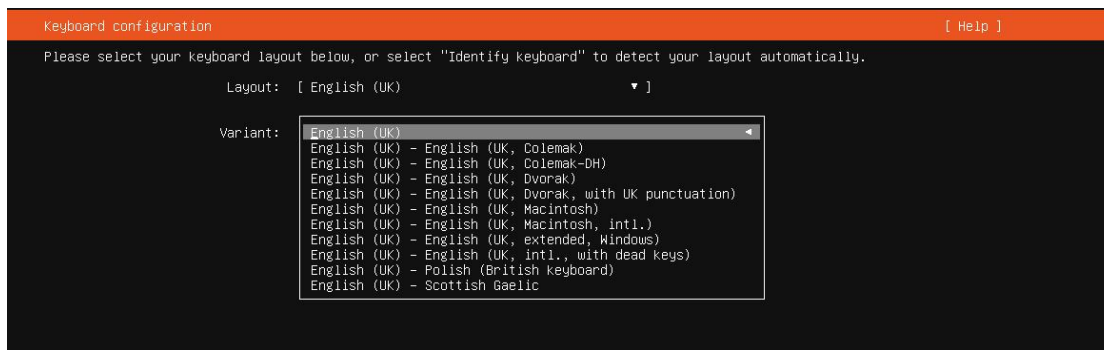
Task 6: Configure the Keyboard Layout

Keyboard Layout



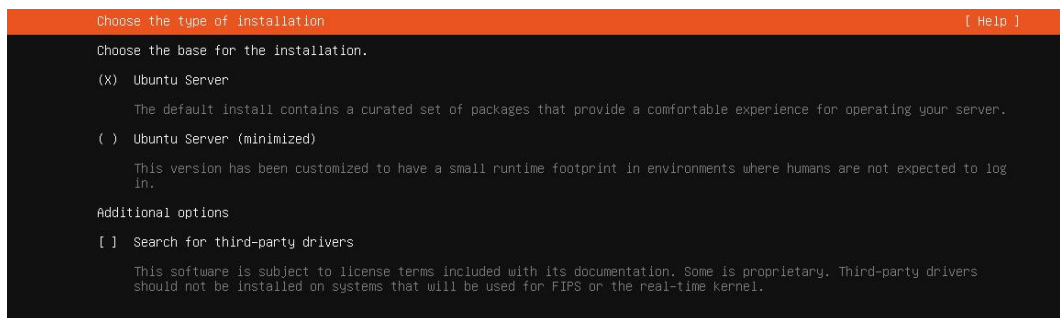
Caption 16: The keyboard layout selection screen of the Ubuntu Server installer.

Keyboard Layout Variant



Caption 17: The keyboard layout variant selection screen, used to fine-tune the chosen layout.

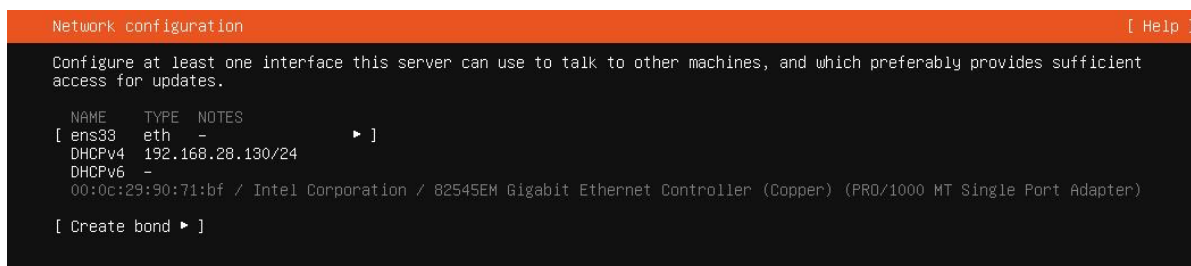
Task 7: Choose the Ubuntu Server Installation Type



Caption 18: The installation type selection screen, offering standard and minimal Ubuntu Server setups.

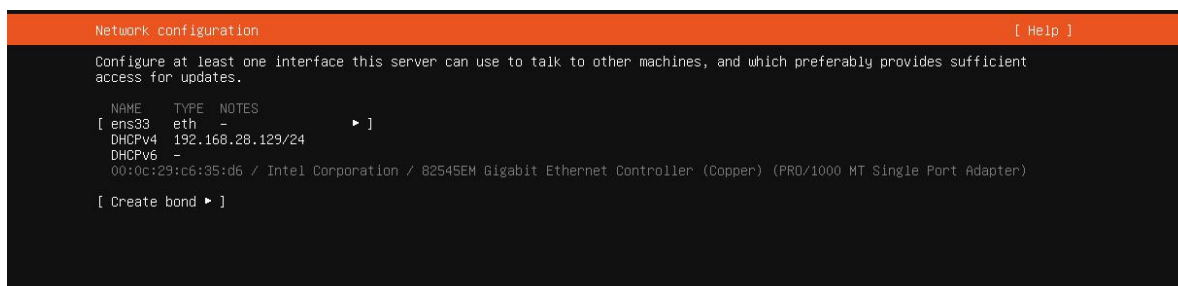
Task 8: Configure the Network

Detected Network Interface



Caption 19: The network interface (ens33) automatically detected by the Ubuntu Server installer.

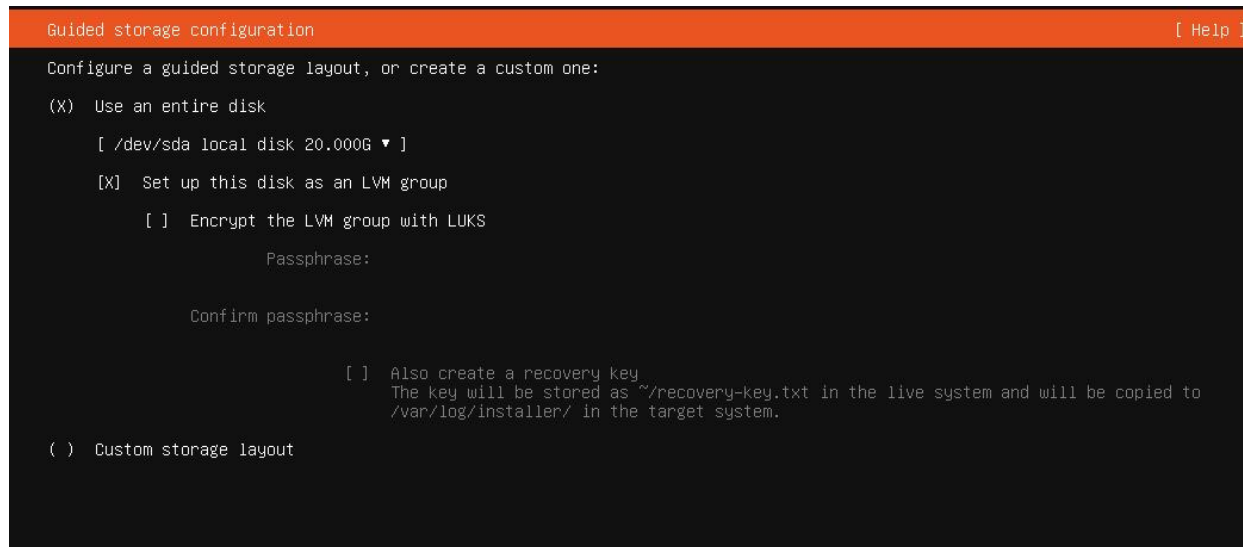
Network Configuration Confirmation



Caption 20: Confirmation that an IP address was assigned to the interface automatically via DHCP.

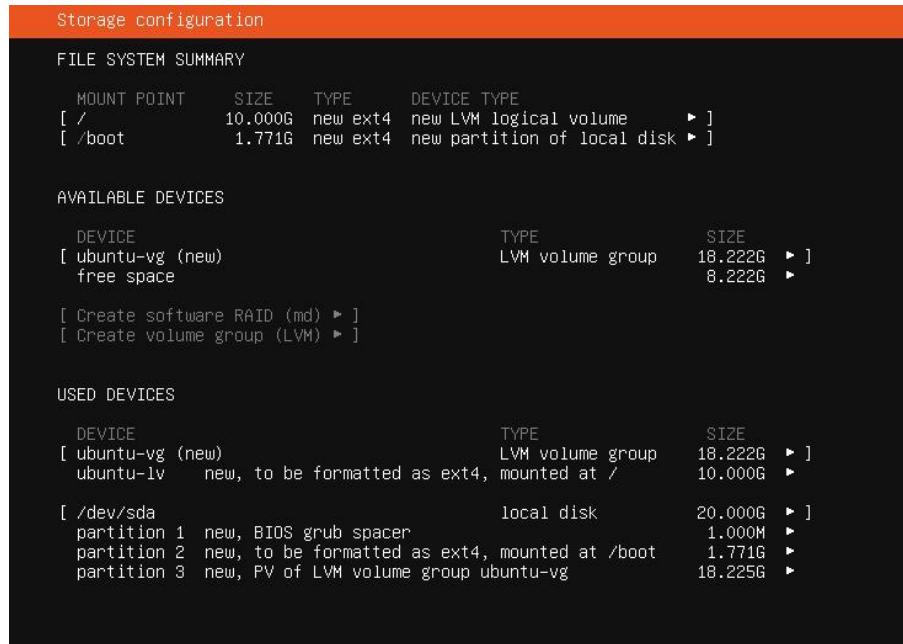
Task 9: Configure Storage

Guided Storage Configuration



Caption 21: The guided storage configuration screen with the "Use an entire disk" option selected.

Storage Configuration Summary



Caption 22: A summary of the file system and storage configuration generated by the installer.

Task 10: Select the Installation Disk

Selected Installation Disk

```

Storage configuration

FILE SYSTEM SUMMARY

  MOUNT POINT      SIZE      TYPE      DEVICE TYPE
[ /                10.000G   new ext4   new LVM logical volume ► ]
[ /boot            1.771G   new ext4   new partition of local disk ► ]

AVAILABLE DEVICES

  DEVICE                                TYPE                                SIZE
[ ubuntu-vg (new)                        LVM volume group                    18.222G ► ]
  free space                             8.222G ► ]

[ Create software RAID (md) ► ]
[ Create volume group (LVM) ► ]

USED DEVICES

  DEVICE                                TYPE                                SIZE
[ ubuntu-vg (new)                        LVM volume group                    18.222G ► ]
  ubuntu-lv    new, to be formatted as ext4, mounted at / 10.000G ► ]

[ /dev/sda                                local disk                          20.000G ► ]
  partition 1  new, BIOS grub spacer                      1.000M ►
  partition 2  new, to be formatted as ext4, mounted at /boot 1.771G ►
  partition 3  new, PV of LVM volume group ubuntu-vg          18.225G ►

```

Caption 23: The VMware virtual disk selected as the target for the Ubuntu Server installation.

Disk Details

```

Storage configuration

FILE SYSTEM SUMMARY

  MOUNT POINT      SIZE      TYPE      DEVICE TYPE
[ /                10.000G   new ext4   new LVM logical volume ► ]
[ /boot            1.771G   new ext4   new partition of local disk ► ]

AVAILABLE DEVICES

  DEVICE                                TYPE                                SIZE
[ ubuntu-vg (new)                        LVM volume group                    18.222G ► ]
  free space                             8.222G ► ]

[ Create software RAID (md) ► ]
[ Create volume group (LVM) ► ]

USED DEVICES

  DEVICE                                TYPE                                SIZE
[ ubuntu-vg (new)                        LVM volume group                    18.222G ► ]
  ubuntu-lv    new, to be formatted as ext4, mounted at / 10.000G ► ]

[ /dev/sda                                local disk                          20.000G ► ]
  partition 1  new, BIOS grub spacer                      1.000M ►
  partition 2  new, to be formatted as ext4, mounted at /boot 1.771G ►
  partition 3  new, PV of LVM volume group ubuntu-vg          18.225G ►

```

Caption 24: The name and size of the selected disk, as reported by the installer.

Task 11: Review the Partitions

```

Storage configuration

FILE SYSTEM SUMMARY

  MOUNT POINT      SIZE    TYPE    DEVICE TYPE
[ /                10.000G new ext4 new LVM logical volume ▶ ]
[ /boot            1.771G new ext4 new partition of local disk ▶ ]

AVAILABLE DEVICES

  DEVICE                                TYPE                                SIZE
[ ubuntu-vg (new)                        LVM volume group                    18.222G ▶ ]
  free space                             8.222G ▶ ]

[ Create software RAID (md) ▶ ]
[ Create volume group (LVM) ▶ ]

USED DEVICES

  DEVICE                                TYPE                                SIZE
[ ubuntu-vg (new)                        LVM volume group                    18.222G ▶ ]
  ubuntu-lv    new, to be formatted as ext4, mounted at / 10.000G ▶ ]

[ /dev/sda                                local disk                          20.000G ▶ ]
  partition 1  new, BIOS grub spacer                      1.000M ▶ ]
  partition 2  new, to be formatted as ext4, mounted at /boot 1.771G ▶ ]
  partition 3  new, PV of LVM volume group ubuntu-vg          18.225G ▶ ]

```

Caption 25: The partition table reviewed and confirmed before proceeding with the installation.

Task 12: Confirm the Storage Changes

Storage Confirmation Warning

```

Confirm destructive action

Selecting Continue below will begin the installation process and
result in the loss of data on the disks selected to be formatted.

You will not be able to return to this or a previous screen once the
installation has started.

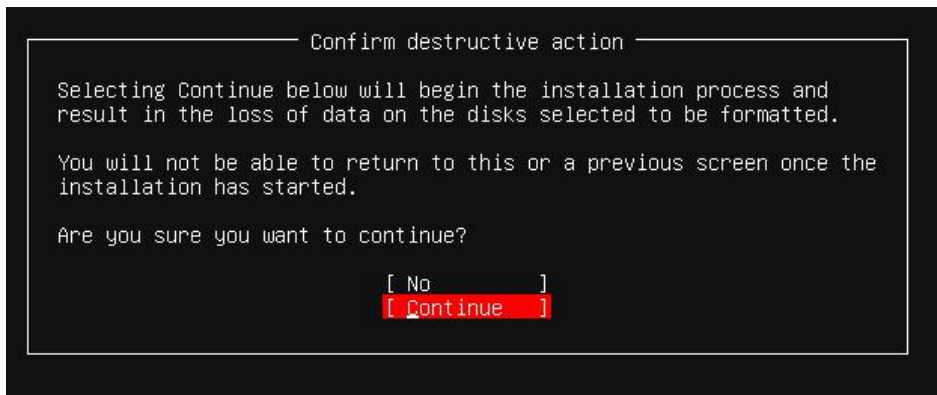
Are you sure you want to continue?

[ No ]
[ Continue ]

```

Caption 26: A warning dialog confirming that continuing will erase all existing data on the selected disk.

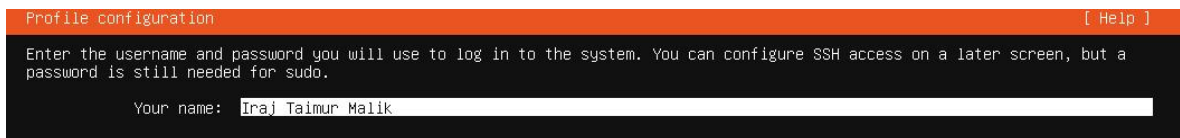
Storage Configuration Confirmed



Caption 27: The storage configuration confirmed, allowing the installation process to proceed.

Task 13: Configure the Ubuntu Profile

Full Name



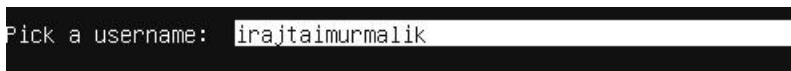
Caption 28: The user's full name entered on the profile configuration screen.

Server Hostname



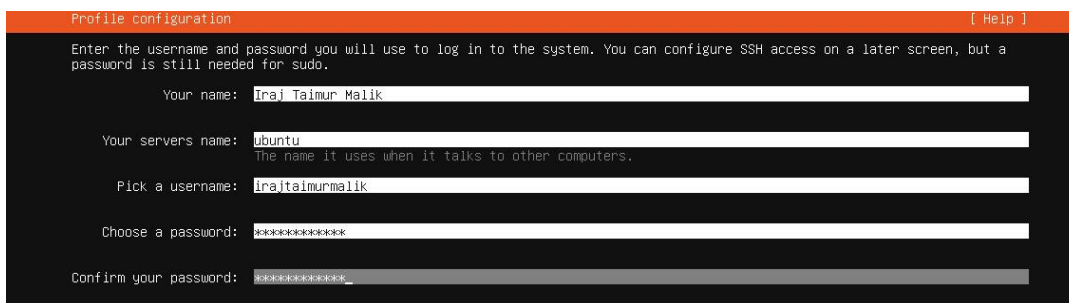
Caption 29: The server's hostname entered during profile configuration.

Username



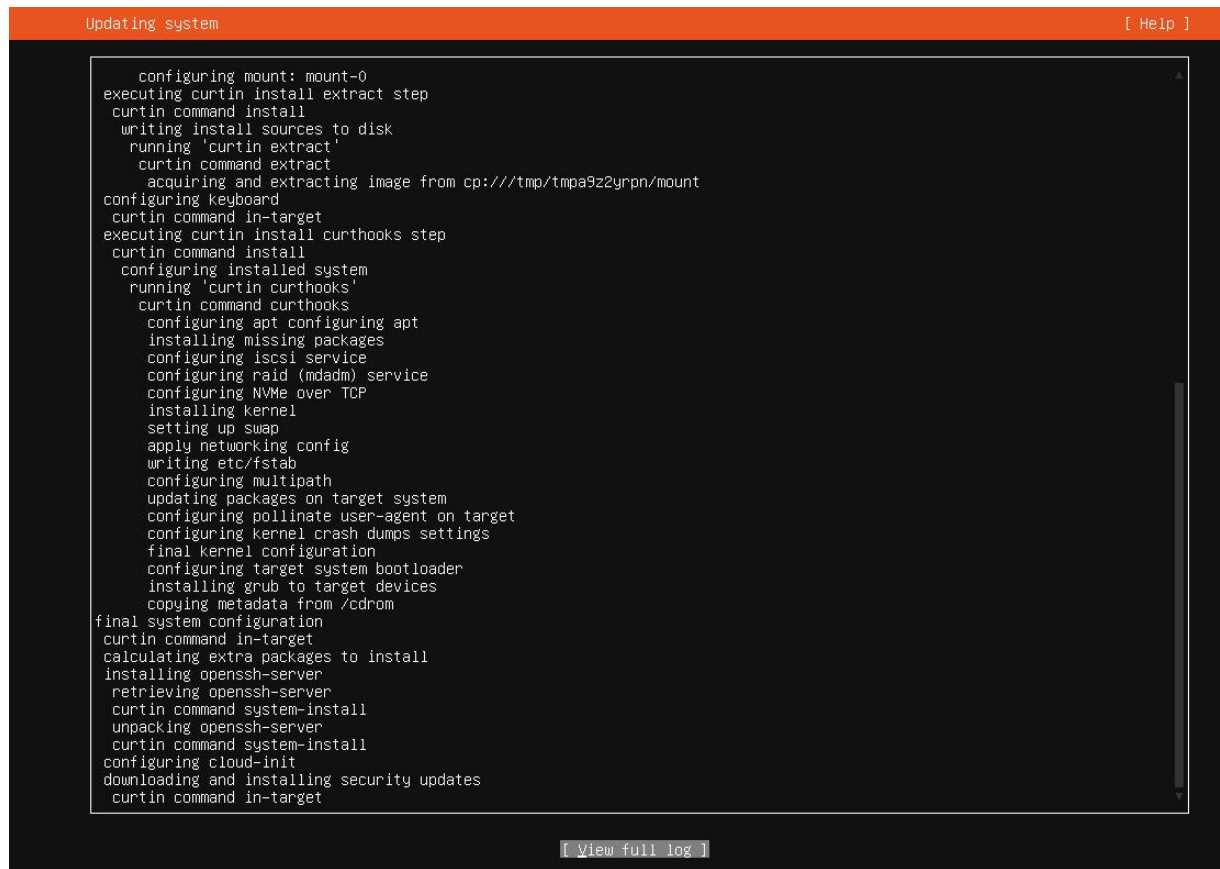
Caption 30: The login username entered during profile configuration.

Profile Review



Caption 31: The completed profile configuration reviewed before continuing with the installation.

Task 14: Complete the Software Installation



```
Updating system [ Help ]

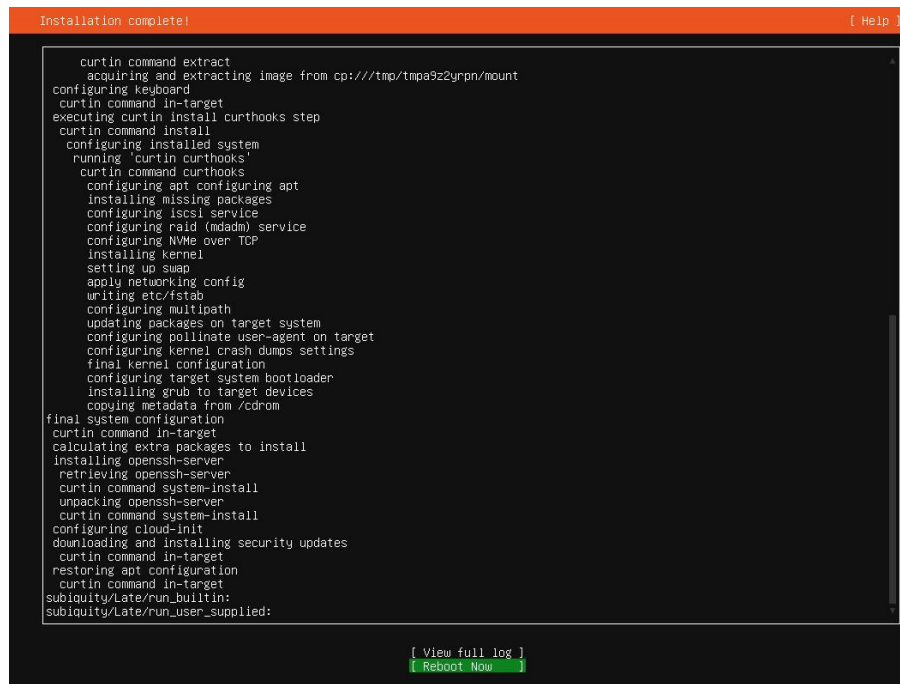
configuring mount: mount-0
executing curtin install extract step
curtin command install
writing install sources to disk
running 'curtin extract'
curtin command extract
acquiring and extracting image from cp:///tmp/tmpa9z2yrpn/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
configuring installed system
running 'curtin curthooks'
curtin command curthooks
configuring apt
configuring apt
installing missing packages
configuring iscsi service
configuring raid (mdadm) service
configuring NVMe over TCP
installing kernel
setting up swap
apply networking config
writing etc/fstab
configuring multipath
updating packages on target system
configuring pollinate user-agent on target
configuring kernel crash dumps settings
final kernel configuration
configuring target system bootloader
installing grub to target devices
copying metadata from /cdrom
final system configuration
curtin command in-target
calculating extra packages to install
installing openssh-server
retrieving openssh-server
curtin command system-install
unpacking openssh-server
curtin command system-install
configuring cloud-init
downloading and installing security updates
curtin command in-target

[ View full log ]
```

Caption 32: The live progress log generated while the Ubuntu Server system files are installed.

Task 15: Reboot and Verify the Ubuntu Identity

Installation Complete



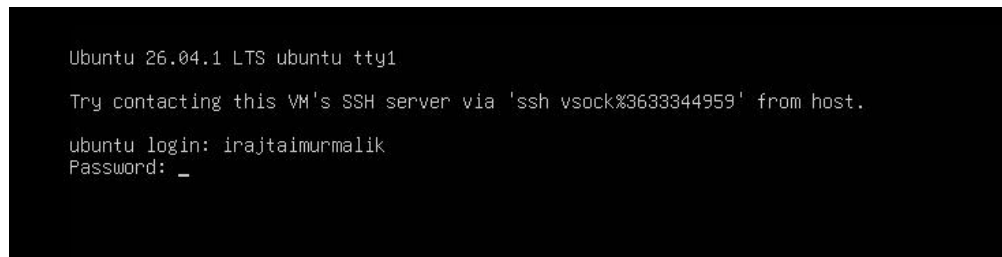
```
Installation complete! [ Help ]

curtin command extract
  acquiring and extracting image from cp:///tmp/tmpa9z2yrpn/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
  configuring installed system
    running 'curtin curthooks'
curtin command curthooks
  configuring apt
  configuring apt
  installing missing packages
  configuring iscsi service
  configuring raid (mdadm) service
  configuring NWE over TCP
  installing kernel
  setting up swap
  apply networking config
  writing etc/fstab
  configuring multipath
  updating packages on target system
  configuring pollinate user-agent on target
  configuring kernel crash dumps settings
  final kernel configuration
  configuring target system bootloader
  installing grub to target devices
  copying metadata from /cdrom
final system configuration
curtin command in-target
calculating extra packages to install
installing openssh-server
retrieving openssh-server
curtin command system-install
unpacking openssh-server
curtin command system-install
configuring cloud-init
downloading and installing security updates
curtin command in-target
restoring apt configuration
curtin command in-target
subiquity/Late/run_builtins:
subiquity/Late/run_user_supplied:

[ View full log ]
[ Reboot Now ]
```

Caption 33: The installation complete screen, offering the option to reboot into the new system.

Ubuntu Login Screen



```
Ubuntu 26.04.1 LTS ubuntu tty1

Try contacting this VM's SSH server via 'ssh vsock%3633344959' from host.

ubuntu login: irajtaimurmalik
Password: _
```

Caption 34: The Ubuntu Server login prompt displayed after the system rebooted.

Terminal Login

```
Ubuntu 26.04.1 LTS ubuntu tty1

Try contacting this VM's SSH server via 'ssh vsock%3633344959' from host.

ubuntu login:
ubuntu login: irajtaimurmalik
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Sep 17 02:52:10 PM UTC 2026

System load:  0.05          Processes:            250
Usage of /:   50.5% of 9.75GB Users logged in:          0
Memory usage: 22%          IPv4 address for ens33: 192.168.28.131
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

13 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

irajtaimurmalik@ubuntu:~$ _
```

Caption 35: A terminal session opened immediately after logging into Ubuntu Server.

Identity Verification

```
irajtaimurmalik@ubuntu:~$ _
```

Caption 36: Command output confirming the identity of the currently logged-in user.

Task 16: Find the Ubuntu Server IP Address

IP Address Command

```
irajtaimurmalik@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:90:71:bf brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c299071bf
    inet 192.168.28.131/24 metric 100 brd 192.168.28.255 scope global dynamic ens33
        valid_lft 1360sec preferred_lft 1360sec
    inet6 fe80::20c:29ff:fe90:71bf/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
irajtaimurmalik@ubuntu:~$ _
```

Caption 37: Output of the `ip addr` command, listing the server's network interfaces.

Server IP Address

```
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:90:71:bf brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c299071bf
    inet 192.168.28.131/24 metric 100 brd 192.168.28.255 scope global dynamic ens33
        valid_lft 1360sec preferred_lft 1360sec
    inet6 fe80::20c:29ff:fe90:71bf/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
```

Caption 38: The Ubuntu Server's assigned IP address, highlighted for reference.

Task 17: Connect to Ubuntu Server from Windows

SSH Command (Windows)

```
Microsoft Windows [Version 10.0.26200.9445]
(c) Microsoft Corporation. All rights reserved.

C:\Users\User>ssh irajtaimurmalik@192.168.28.131
The authenticity of host '192.168.28.131 (192.168.28.131)' can't be established.
ED25519 key fingerprint is SHA256:vP9WQdVK18r526n0E12UhmqBIbjsq19L7uNpCqDah/Q.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? |
```

Caption 39: The SSH command entered from a Windows terminal to connect to the Ubuntu Server.

SSH Fingerprint

```
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.28.131' (ED25519) to the list of known hosts.
irajtaimurmalik@192.168.28.131's password: |
```

Caption 40: The SSH host key fingerprint confirmation prompt shown on first connection.

SSH Login

```
irajtaimurmalik@192.168.28.131's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Sep 17 03:07:48 PM UTC 2026

System load:  0.02               Processes:            220
Usage of /:   50.6% of 9.75GB    Users logged in:     0
Memory usage: 22%               IPv4 address for ens33: 192.168.28.131
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

13 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

irajtaimurmalik@ubuntu:~$ |
```

Caption 41: A successful SSH login session established with the Ubuntu Server.

SSH Identity Verification

```
irajtaimurmalik@ubuntu:~$
```

Caption 42: Command output verifying the user's identity after connecting via SSH.